

Capture the Flag 4: The Student List

Goal: Build a scrollable student directory using Dart Lists and `ListView.builder`, replacing manually duplicated widgets with dynamically generated UI.

Instructor Note

These activities require your own problem-solving and research skills.

DO NOT use AI to generate the code for you.

You may use AI to:

- Research what a Dart/Flutter feature does
- Understand syntax
- Explain an error (not fix your error)
- Clarify documentation

You may NOT ask AI to generate the solution/code for a flag.

🚩 Flag 0 — Create Your Feature Branch

Objective: Create a feature branch before building your student list.

Create:

`feature/student-list`

Screenshot:

- Terminal showing your current branch

```
C:\Users\johnf\Downloads\ggs\my_first_flutter_app>git branch
feature/data-ui
* feature/student-list
feature/widgets
main
```

🚩 Flag 1 — Build Your Student Data

Objective: Store multiple students inside a Dart `List`.

Research what a Dart `List` is and why it is useful when handling multiple pieces of similar data.

Instead of creating one student at a time with separate variables, create a collection that stores at least five students.

Each student should contain information such as:

- Image
- Name
- Course
- Year Level
- Age
- Hobby

Your data should exist separately from your UI widgets.

Screenshot:

Code:

- Your student list
- At least five student entries

```
8  class Student {
9      final String image;
10     final String name;
11     final String course;
12     final String yearLevel;
13     final int age;
14     final String hobby;
15
16     const Student({
17         required this.image,
18         required this.name,
19         required this.course,
20         required this.yearLevel,
21         required this.age,
22         required this.hobby,
23     });
24 }
```

```
28  const List<Student> students = [
29      Student(
30          image: "assets/profile2.jpg",
31          name: "Ivan Villareal",
32          course: "BSIT",
33          yearLevel: "3rd Year",
34          age: 24,
35          hobby: "Photography, Videography, & Video Editing",
36      ),
37
38      Student(
39          image: "assets/Profile1.jpg",
40          name: "Mannes Artajo",
41          course: "BSIT",
42          yearLevel: "2nd Year",
43          age: 21,
44          hobby: "Reading"
```

```

47     Student(
48         image: "assets/profile3.jpg",
49         name: "Wilken Montebon",
50         course: "BSIT",
51         yearLevel: "1st Year",
52         age: 20,
53         hobby: "Basketball",
54     ),
55
56     Student(
57         image: "assets/profile4.png",
58         name: "John Kevin Villacorte",
59         course: "BSIT",
60         yearLevel: "4th Year",
61         age: 23,
62         hobby: "Drawing",
63     ),
64
65     Student(
66         image: "assets/profile5.jpg",
67         name: "Keith Francheska Lopez",
68         course: "BSIT",
69         yearLevel: "3rd Year",
70         age: 22,
71         hobby: "Gaming",
72     ),
73 ];

```

🚩 Flag 2 — The Copy-Paste Trap

You already know how to create a Student Card.

You created five student cards manually.

You duplicated the same card design until all five students appear.

Stop.

Imagine having:

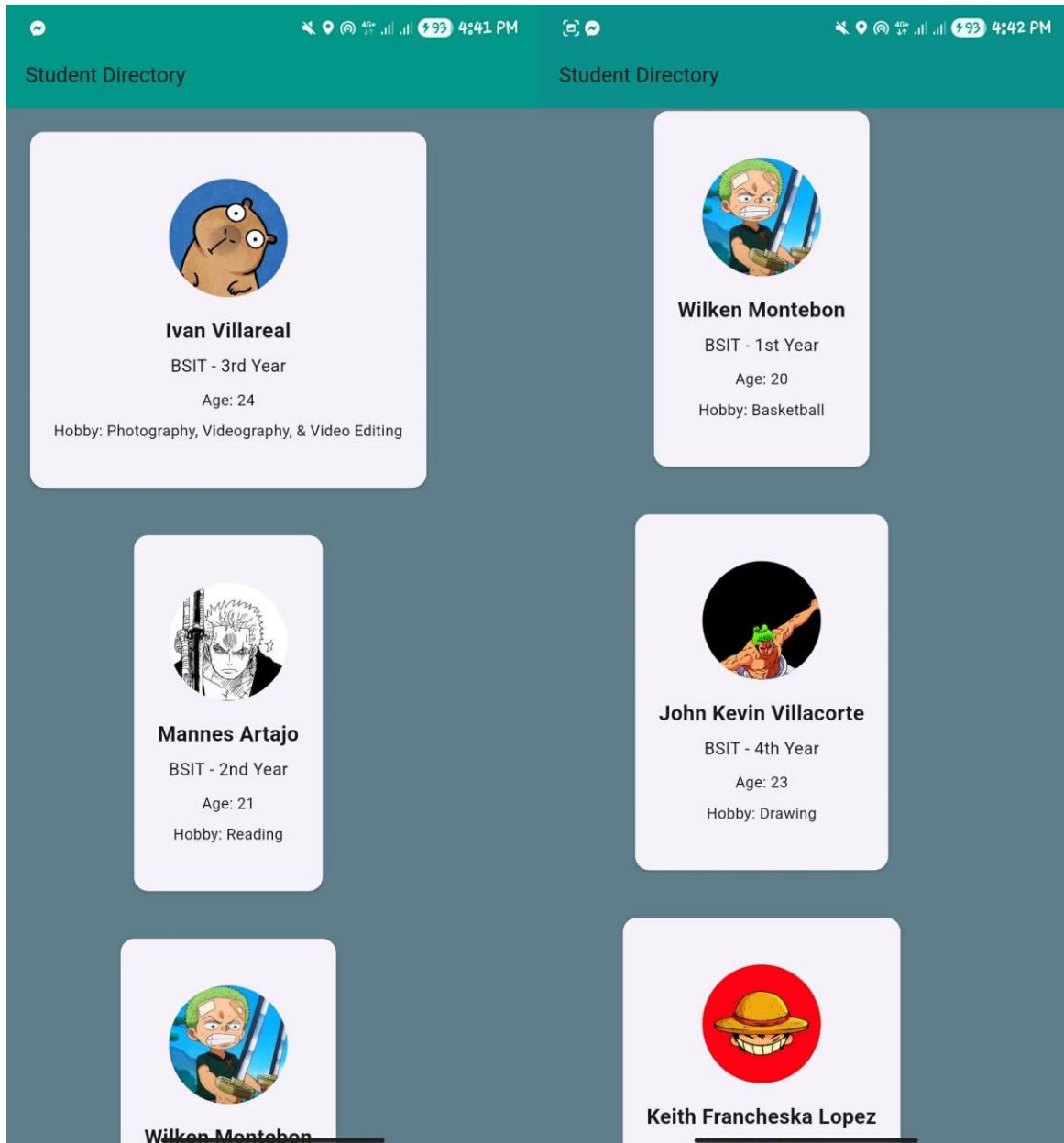
- 50 students.
- 500 students.
- 5,000 students.

Would copying the same widget over and over still be practical?

This flag is intentionally making you feel the problem before solving it.

Screenshot:

- Your five manually created student cards (do not change your past code yet)





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Student Directory



Wilken Montebon

BSIT - 1st Year

Age: 20

Hobby: Basketball



John Kevin Villacorte

BSIT - 4th Year

Age: 23

Hobby: Drawing



Keith Francheska Lopez

BSIT - 3rd Year

Age: 22

Hobby: Gaming

```

94 Card(
95   margin: const EdgeInsets.all(20),
96   child: SizedBox(
97     height: 300,
98     child: Padding(
99       padding: const EdgeInsets.all(20),
100      child: Column(
101        mainAxisAlignment: MainAxisAlignment.center,
102        children: [
103          CircleAvatar(
104            radius: 50,
105            backgroundImage:
106              AssetImage(students[0].image),
107          ), CircleAvatar

```

```

148 Card(
149   margin: const EdgeInsets.all(20),
150   child: SizedBox(
151     height: 300,
152     child: Padding(
153       padding: const EdgeInsets.all(20),
154       child: Column(
155         mainAxisAlignment: MainAxisAlignment.center,
156         children: [
157           CircleAvatar(
158             radius: 50,
159             backgroundImage:
160               AssetImage(students[1].image),
161           ), CircleAvatar
162

```

```

203   margin: const EdgeInsets.all(20),
204   child: SizedBox(
205     height: 300,
206     child: Padding(
207       padding: const EdgeInsets.all(20),
208       child: Column(
209         mainAxisAlignment: MainAxisAlignment.center,
210         children: [
211           CircleAvatar(
212             radius: 50,
213             backgroundImage:
214               AssetImage(students[2].image),
215           ), CircleAvatar
216
217   const SizedBox(height: 15),

```

```

256 Card(
257   margin: const EdgeInsets.all(20),
258   child: SizedBox(
259     height: 300,
260     child: Padding(
261       padding: const EdgeInsets.all(20),
262       child: Column(
263         mainAxisAlignment: MainAxisAlignment.center,
264         children: [
265           CircleAvatar(
266             radius: 50,
267             backgroundImage:
268               AssetImage(students[3].image),
269           ), CircleAvatar
270
271           const SizedBox(height: 15),
272
310 Card(
311   margin: const EdgeInsets.all(20),
312   child: SizedBox(
313     height: 300,
314     child: Padding(
315       padding: const EdgeInsets.all(20),
316       child: Column(
317         mainAxisAlignment: MainAxisAlignment.center,
318         children: [
319           CircleAvatar(
320             radius: 50,
321             backgroundImage:
322               AssetImage(students[4].image),
323           ), CircleAvatar
324
325           const SizedBox(height: 15),

```

🚩 Flag 3 — Discover **ListView**

Objective: Make your student directory scrollable using **ListView**.

Research Flutter's **ListView**.

Replace your previous layout with a scrolling layout so users can browse through all of your student cards naturally.

Your application should now allow continuous vertical scrolling.

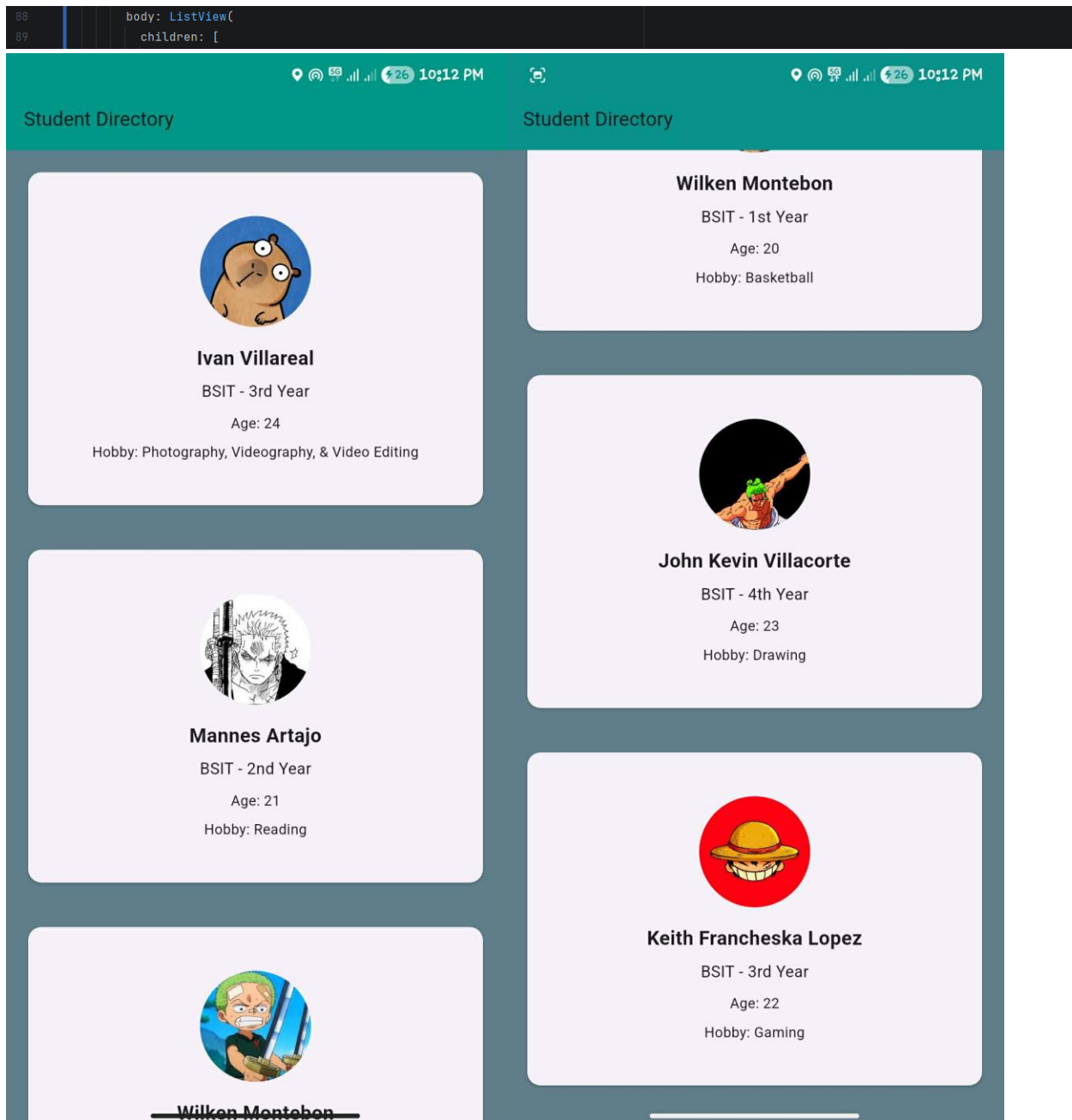
Test

The user should be able to:

- Scroll from the first student
- Continue scrolling
- Reach the last student without resizing the screen

Screenshot:

- `ListView` in your code
- Top of the application
- Bottom portion showing later students



🚩 Flag 4 — Unlock `ListView.builder`

Objective: Generate student cards automatically using `ListView.builder`.

Research `ListView.builder`.

Instead of writing five separate Student Cards, generate them automatically from your student list.

Your UI should now create one card for each student in your list.

Test

Add one new student to your list.

Without creating another card manually, your application should display the new student automatically.

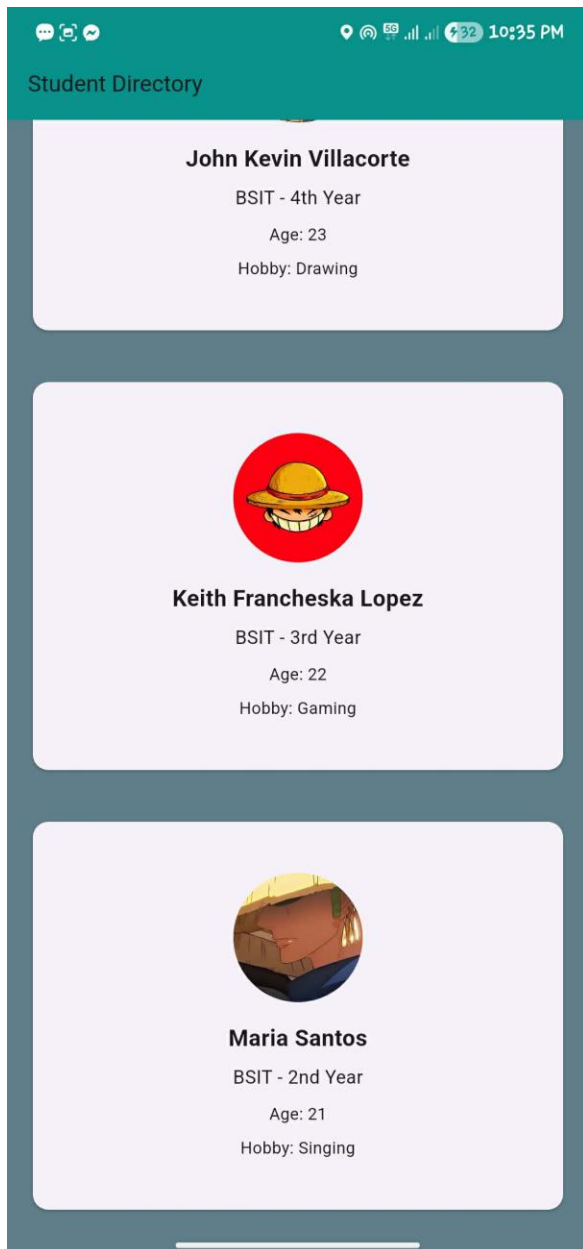
Screenshot:

- `ListView.builder` in your code
- Student list
- Newly generated student appearing in the app

```
103   body: ListView.builder(  
104     itemCount: students.length,  
105  
106     itemBuilder: (context, index) {  
107  
108       final student = students[index];  
109  
110       return Card(  
  
28   const List<Student> students = [  
29     Student(  
30       image: "assets/profile2.jpg",  
31       name: "Ivan Villareal",  
32       course: "BSIT",  
33       yearLevel: "3rd Year",  
34       age: 24,  
35       hobby: "Photography, Videography, & Video Editing",  
36     ),  
37  
38     Student(  
39       image: "assets/Profile1.jpg",  
40       name: "Mannes Artajo",  
41       course: "BSIT",  
42       yearLevel: "2nd Year",  
43       age: 21,  
44       hobby: "Reading",  
45     ),  
  ],  
);
```

```
47 Student(  
48     image: "assets/profile3.jpg",  
49     name: "Wilken Montebon",  
50     course: "BSIT",  
51     yearLevel: "1st Year",  
52     age: 20,  
53     hobby: "Basketball",  
54 ),  
55  
56 Student(  
57     image: "assets/profile4.png",  
58     name: "John Kevin Villacorte",  
59     course: "BSIT",  
60     yearLevel: "4th Year",  
61     age: 23,  
62     hobby: "Drawing",  
63 ),
```

```
65 Student(  
66     image: "assets/profile5.jpg",  
67     name: "Keith Francheska Lopez",  
68     course: "BSIT",  
69     yearLevel: "3rd Year",  
70     age: 22,  
71     hobby: "Gaming",  
72 ),  
73  
74 // FLAG 4 - New Student  
75 Student(  
76     image: "assets/Profile6.jpg",  
77     name: "Maria Santos",  
78     course: "BSIT",  
79     yearLevel: "2nd Year",  
80     age: 21,  
81     hobby: "Singing",  
82 ),  
83 ];
```



🚩 Flag 5 — Every Student Deserves an ID

Objective: Expand your student records with additional information.

Real student directories contain more than just names.

Add at least three new fields to every student.

Examples:

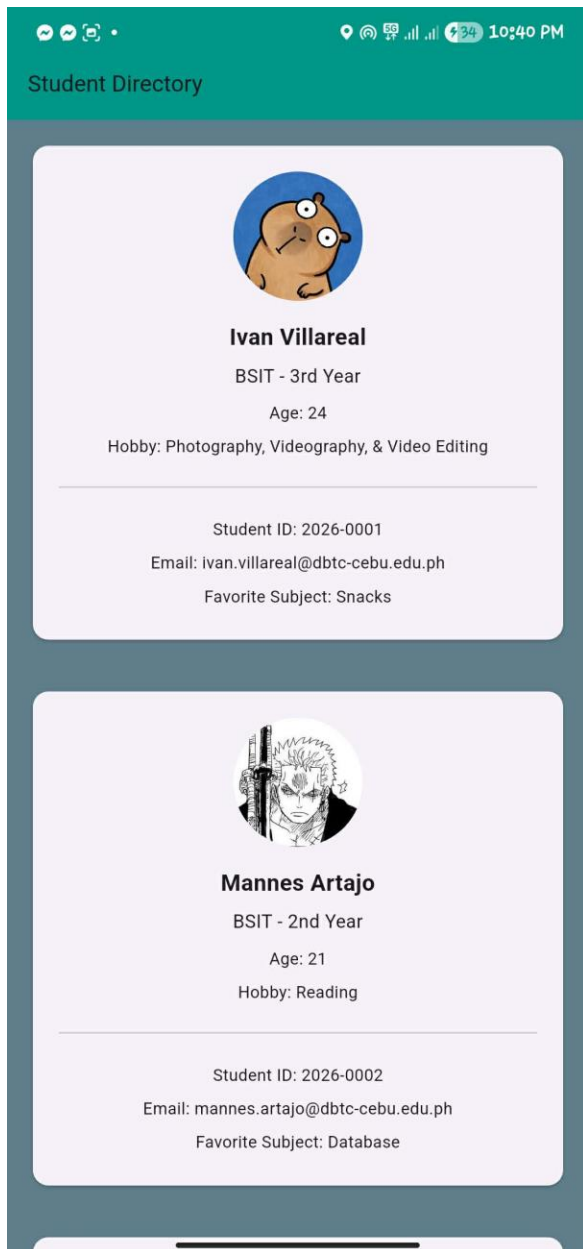
- Student ID
- Email
- Favorite Subject

Update your Student Card so the new information appears consistently across every generated card.

Screenshot:

- Updated student data
- Student Card displaying the additional fields

```
39 Student(  
40     image: "assets/profile2.jpg",  
41     name: "Ivan Villareal",  
42     course: "BSIT",  
43     yearLevel: "3rd Year",  
44     age: 24,  
45     hobby: "Photography, Videography, & Video Editing",  
46  
47     // FLAG 5  
48     studentId: "2026-0001",  
49     email: "ivan.villareal@dbtc-cebu.edu.ph",  
50     favoriteSubject: "Snacks",  
51 ),
```



🚩 Flag 6 — Empty Classroom

What happens when there are no students?

Objective: Handle an empty `List` gracefully.

Imagine your app just connected to a database.

The database returns:

[]

No students.

What should the user see?

Instead of displaying a broken or confusing screen, research how Flutter can display a different widget when a condition is met.

Create an empty state that clearly communicates that there are currently no students available.

Examples:

- "No students found."
- "Student list is empty."

Test

- Display your student list normally.
- Then temporarily make your list empty.
- Your application should display the empty state instead of student cards.

Screenshot:

- Empty student list
- Empty-state UI
- Application still functioning correctly

```
35     const List<Student> students = [];  
36
```

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Student Directory

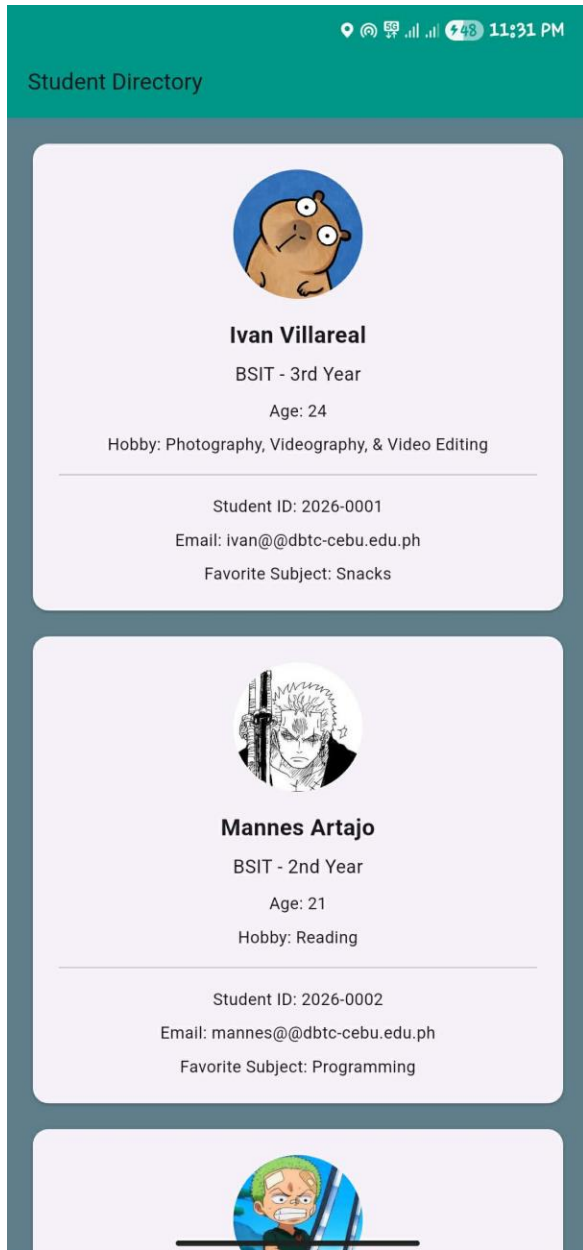


No students found.

The student list is currently empty.

```
54     body: students.isEmpty
55       ? const Center(
56         child: Column(
57           mainAxisAlignment: MainAxisAlignment.center,
58           children: [
59             Icon(
60               Icons.people_outline,
61               color: Colors.white,
62               size: 70,
63             ), Icon
```

```
35   const List<Student> students = [
36     Student(
37       image: "assets/profile2.jpg",
38       name: "Ivan Villareal",
39       course: "BSIT",
40       yearLevel: "3rd Year",
41       age: 24,
42       hobby: "Photography, Videography, & Video Editing",
43       studentId: "2026-0001",
44       email: "ivan@ddbtc-cebu.edu.ph",
45       favoriteSubject: "Snacks",
46     ),|
47
48     Student(
49       image: "assets/Profile1.jpg",
```



🚩 Flag 7 — Sort the Roll Call

Objective: Change the order of your student list.

Class lists are rarely displayed randomly.

Research how Dart can rearrange items inside a [List](#).

Sort your students alphabetically by name.

Test

Before sorting:

- Maria
- John
- Aaron
- Sophia

After sorting:

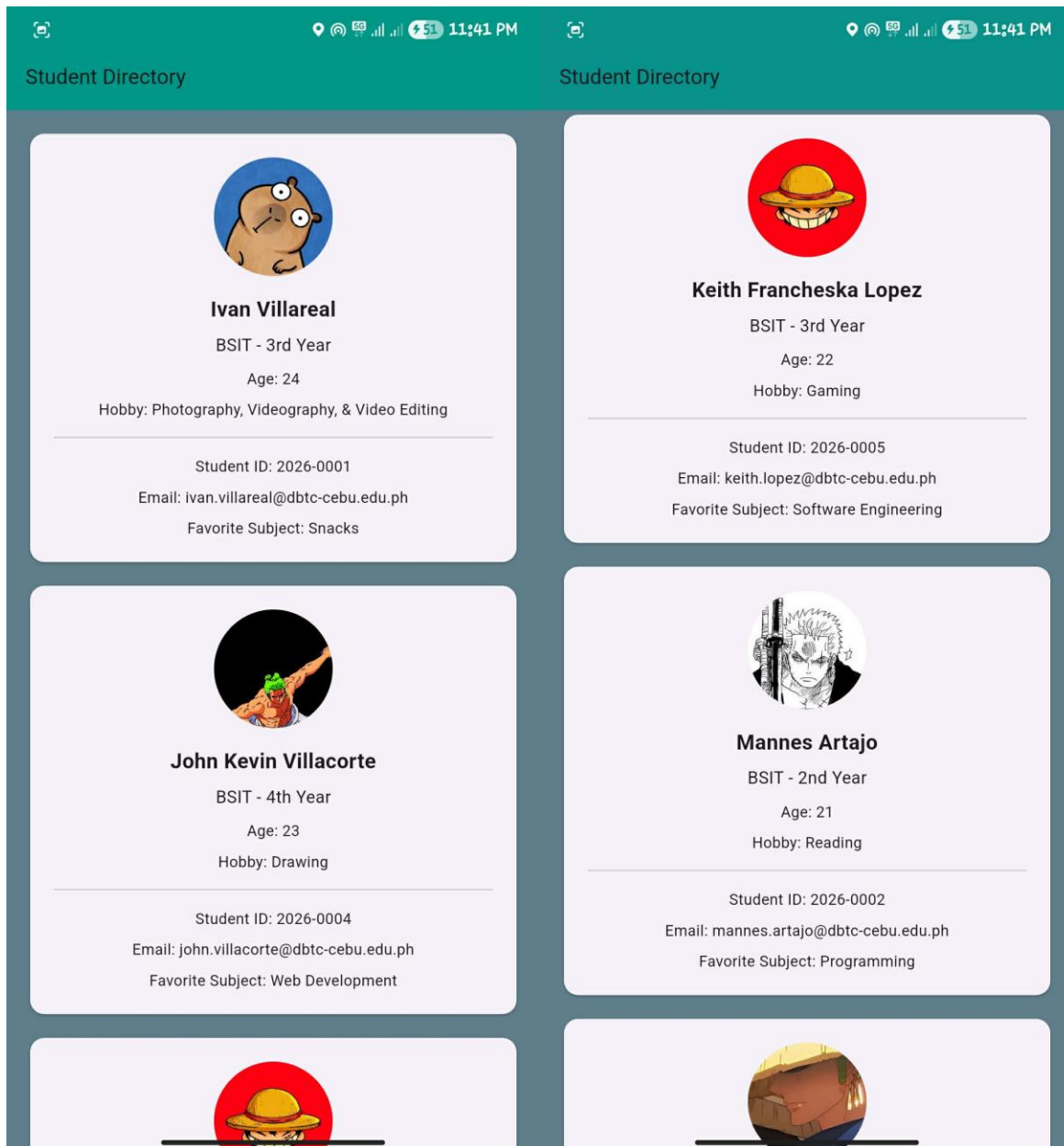
- Aaron
- John
- Maria
- Sophia

Your UI should immediately reflect the new order.

Screenshot:

- Student list before or after sorting
- Application displaying the sorted order

```
3      void main() {
4          // FLAG 7 - Sort students alphabetically before running the app
5          students.sort(
6              (a, b) => a.name.toLowerCase().compareTo(
7                  b.name.toLowerCase(),
8              ),
9          );
43     List<Student> students = [
```



🚩 Flag 8 — Save Your Progress

Objective: Commit and push your completed Student List CTF.

Use an appropriate commit message following proper Git standards.

Branch:

`feature/student-list`

Screenshot:

- Successful commit
- Current branch
- Terminal showing the commit

```
C:\Users\johnf\Downloads\ggs>git branch
  feature/data-ui
* feature/student-list
  feature/widgets
  main

C:\Users\johnf\Downloads\ggs>git add .

C:\Users\johnf\Downloads\ggs>git commit -m
error: switch '-m' requires a value

C:\Users\johnf\Downloads\ggs>git commit -m "Feat: Added PDF File"
[feature/student-list 45396aa] Feat: Added PDF File
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 IT314AB Capture The Flag 4 - Villareal, Ivan Vincent C..pdf

C:\Users\johnf\Downloads\ggs>git push origin feature/student-list
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Compressing objects: 100% (4/4), done.
Successfully pushed.

C:\Users\johnf\Downloads\ggs\my_first_flutter_app>git status
On branch feature/student-list
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   assets/Profile6.jpg
    modified:   lib/main.dart

C:\Users\johnf\Downloads\ggs\my_first_flutter_app>git commit -m "Feat: complete student list CTF"
[feature/student-list e85aa03] Feat: complete student list CTF
 2 files changed, 198 insertions(+), 169 deletions(-)
 create mode 100644 my_first_flutter_app/assets/Profile6.jpg

C:\Users\johnf\Downloads\ggs\my_first_flutter_app>git push origin feature/student-list
Enumerating objects: 12, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 12 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (7/7), 64.72 KiB | 21.57 MiB/s, done.
Total 7 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), completed with 4 local objects.
remote:
remote: Create a pull request for 'feature/student-list' on GitHub by visiting:
remote:   https://github.com/ivan013102/IT314AB_VILLAREAL/pull/new/feature/student-list
remote:
To https://github.com/ivan013102/IT314AB_VILLAREAL.git
 * [new branch]      feature/student-list -> feature/student-list

C:\Users\johnf\Downloads\ggs\my_first_flutter_app>
```